Universal Dependencies for Hittite

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The project aims to develop a Universal Dependencies (UD) treebank for Hittite, the most ancient Indo-European language attested in writing (cuneiform, II millenium BC). As a case study, the letter of the Hittite Queen Puduheba (CTH 176, KUB 21.38) was parsed and annotated for UD, containing 1,254 words (193 clauses). The previous research (Inglese et al. 2018) yielded another 313 clauses pre-annotated in the dependency framework.
The development of Hittite UD treebanks naturally meets certain challenges that we discuss below, such as annotation of clitic chains, complex sentences, null auxiliaries. This work builds on previous projects [1], [2], [3], [4].

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Introduction

Universal Dependencies (US) is a framework for consistent annotation of grammar across different human languages. Hittite, however, still lacks a proper UD annotation. For now, there is only one treebank of Hittite built upon some data retrival from the classical grammar of Hittite [5]. This project is an on-going work for developing a consistent treebank of the Hittite letters and instructions.
As a first approach, the Letter of the Queen Puduheba to the Egyptian pharaoh Ramses II has been chosen - as one that is long enough and better preserved than most of the other letters and instructions from the Hittite Kingdom.

The King Hattusili III and his wife Puduheba

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Challenges

There are several challenges in the annotation of the Hittite UD. To mention a few:

- Hittite **correlatives**: a part of the complex sentence or an independent clause?
- Hittite **clitic chains**: independent wordforms or one part together with the host?
- Hittite **lemmatization**: why on Earth did they write in three languages, Hittite, Akkadian and Sumerian?

Methodology

The corpus of texts includes Hittite letters, as published by [6] and represented in [2]. Files are being annotated in CONLL-U format. Visualization of the treebank: ArboratorGrew ([7], [8]).

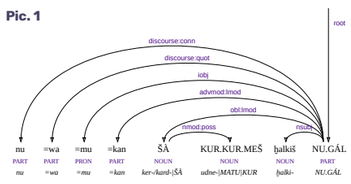
Annotated files and screenshots of generated trees can be downloaded from GitHub: https://github.com/mashenkeisraeli/hittite_ud

Examples

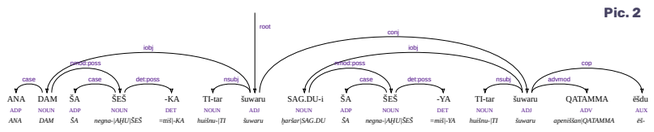
1. *nu=wa=mu=kan* ŠÀ KUR.KUR.MEŠ [*ḫal*]*kiš* NU.GÁL "In my lands I do not even have barley" (Pic. 1.)
2. [*ANA*] DAM *ŠA* ŠEŠ-KA TI-tar šuwaru SAG.DU-i *ŠA* [ŠEŠ-*YA* TI-tar šuwaru *QATAMMA* ēšdu "The wife of your brother (i.e., Puduḫepa, the wife of Ḫattušili) enjoys full life. May the person of my brother likewise enjoy full life!" (reconstruction by Hoffner 2009:282) (Pic. 2.)
3. ŠEŠ-*YA=ma=mu kuit kiššan TÀŠPUR*, NIN-*YA=wa=mu IŠPUR* DUMU.MUNUS=*wa=ta peḫḫi* "My brother, as you wrote to me in this manner, my sister wrote to me: 'I will give my daughter to you.'" (Pic. 3.)
4. *ANA* ŠEŠ-*YA ammel aššulan ammel unuwašḫan uppaḫḫun* "I have sent my greetings (and) my ornaments to my brother"
5. *ammuk=ma ANA* ŠEŠ-*YA kuin* DUMU.MUNUS *nepišaš* KI-*ašš=a peḫḫi n= an =kan kuedani ḫandami* "The daughter of heaven and earth whom I will give to my brother–to whom should I compare her?" (Pic. 4.)
6. *nu=wa* NAM.RA.MEŠ *kuin* GU_4.MEŠ UDU.ḪI.A *pē ḫarkanzi … nu=war=an=kan arḫa daškandu … nu=war=an parnawiškandu* "The captives, cattle, and sheep that they are holding – let them be taken away and let them be housed." (Pic. 5.)

Clitic chains are separate syntactic words

Pic. 1

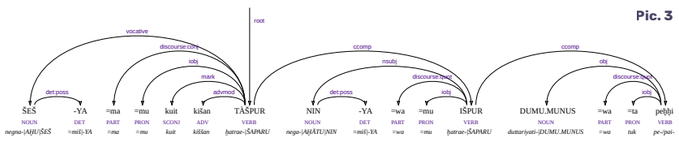


Pic. 2



Reported discourse: ccomp or parataxis

Pic. 3



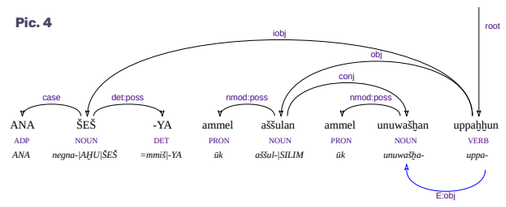
Three languages in one LEMMA

A Sumerogram or Akkadogram spells a Hittite word. The project records all three and puts the Hittite lexeme first, separated by vertical bars, with the written realisation marked in FEATS (Pic. 2)

LEMMA = maḫḫan|GIM ‘when’
LEMMA = negna-|AḪU|ŠEŠ ‘brother’
FEATS = Language=Sum / Language=Akk

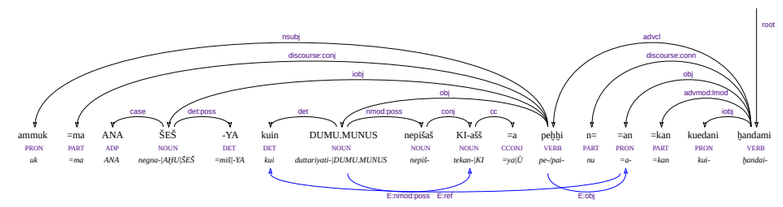
Coordination and Akkadographic prepositions

Pic. 4

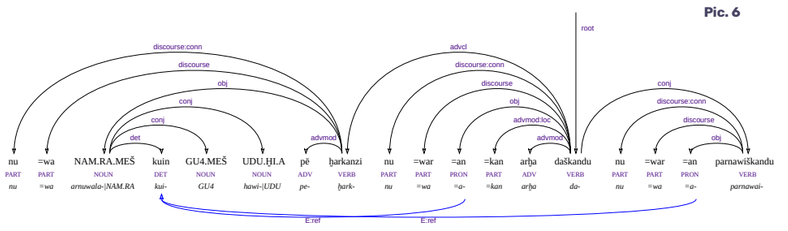


Correlatives in the enhanced graph

Pic. 5



Pic. 6



The seal of Hattusili III and Queen Puduheba

Conclusion

For the time being the corpus of Hittite clauses consists of 105 texts, 2,197 clauses, which include approximately 10,000 words. The work is ongoing, with a promising perspective to eventually create the first Hittite treebank based on the language of real documents.

The results of the UD annotation for the Hittite letters have allowed for the development of new guidelines for Hittite annotation without requiring many changes to the basic UD guidelines. This will facilitate the easier inclusion of Hittite into the Universal Dependencies system and in general will open Hittite texts to automated typological comparison.

References

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